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# A Real-Time Framework for Wrong-Way Driving Detection Using YOLOv9 and Multi-Object Tracking

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**Abstract:** Wrong-way driving (WWD) is one of the most dangerous traffic behaviors, which mainly causes head-on collisions resulting in death. Traditional methods of detection, such as loop detectors and manual surveillance, are often inadequate due to high costs, limited coverage, and delayed response times. In this paper, we present a new real-time computer vision-based framework for automatic detection of WWD instances. The system uses the latest You Only Look Once version 9 (YOLOv9) object detection model for strong and fast vehicle identification. Additionally, a multi-object tracking algorithm is used, which allows the system to keep track of the vehicles' identities across the video frames. The fundamental part of our approach is the arrangement of consecutive virtual detection zones on the road; a vehicle is accused of a WWD violation if it moves through these zones in the wrong order. Our experimental results show that the framework can be a practical and efficient method for obtaining high accuracy and real-time performance. This system presents great promise for practical use after conducting additional experiments with longitudinal and multi-camera models. Besides, it is a cheap and handy method of making roads safer on highways and city streets.

**Keywords:** Wrong-way driving; Intelligent transportation systems; Computer vision; YOLOv9; Object tracking; Deep learning; Traffic safety

## 1 Introduction

Traffic accidents continue to be among the top causes of death and injury that can be avoided worldwide. Among these, wrong-way driving (WWD) incidents are particularly dangerous and are a major cause of serious violations [1]. A WWD situation occurs when a vehicle is driven against the direction of the traffic flow. Such cases almost always lead to direct head-on collisions at high speeds with outcomes that are extremely fatal and injurious due to the severity of the impact [2]. A WWD event is chaotic and hence poses a very daunting challenge not only to law enforcement but also to the other road users. This is why there is a need to come up with highly sophisticated automated systems that can detect and give warnings in time. Most traffic violation detection systems today depend quite a lot on inductive loop detectors, which are essentially wire coils laid down in the road surface [3]. It is not to dispute that these kinds of detectors are capable of properly recording vehicle numbers and even estimating their speeds. On the contrary, their disadvantage is that they are fairly expensive not only for the initial installation but also if you want to keep them running in good order; And, they are limited to a single point only. What is more, they cannot differentiate individual vehicles over a distance, which is an essential feature for something like WWD detection. Considering this, the issue of looking for clever and versatile solutions has been brought up. The widespread use of high-definition (HD) video cameras on main roads is a great source of opportunity for intelligent traffic control and analysis [4]. Automated traffic analysis from video streams is a very achievable task by leveraging computer vision, which can extract comprehensive details like vehicle type, speed, route, and even license plate information. Armed with such data, one can create sophisticated systems that can identify complex traffic violations like WWD without human intervention in real time. Incorporating computer vision into smart transportation systems (intelligent transportation system, ITS) has remained a very significant area of research over the years [5]. At the start, the systems simply applied elementary image processing techniques, e.g., background subtraction, frame differencing, and feature extraction via Histogram of Oriented Gradients (HOG) and Support

Vector Machines (SVM) classifiers [6]. Although these methods provided solutions to automated traffic monitoring, they largely faltered in scenarios of extreme lighting changes, weather alterations, and occlusions.

The emergence of deep learning, especially convolutional neural networks (CNNs), has transformed the domain of object detection and has become the go-to method for vehicle detection [7]. Two-stage detectors such as the Region-based convolutional neural network (R-CNN) family, including Faster R-CNN, which are known for their high accuracy, can be quite demanding in terms of computation; thus, they are generally not considered the best option for real-time processing of video streams unless there is significant hardware acceleration support [8].

This drawback was the main reason behind the creation of single-stage detectors. Single-stage detectors like the You Only Look Once (YOLO) series and the Single Shot MultiBox Detector (SSD) focus on speed; thus, they are very appropriate for real-time uses [9]. The YOLO model treats object detection as a single regression task where, in one go, it predicts the locations of the bounding boxes and the class probabilities directly from the entire image. This idea has sparked several follow-ups, with each newer version (e.g., YOLOv3, YOLOv5, YOLOv8) providing a better compromise between speed and accuracy, hence becoming the preferred choice for traffic monitoring systems [10]. In recent years, WWD detection systems that have been developed still face a major theoretical problem between robustness and computational efficiency. For example, very powerful spatio-temporal graph techniques are almost immune to occlusion, but they require a lot of computing, which makes running these systems in real-time impossible. However, very quick methods like virtual line crossing are computationally efficient, but they are easily broken and lead to a number of false positives (FP) when there are lane changes or partial occlusions. Because of this, there is a significant research gap that is left unfilled: the desire for a trajectory processing approach that is resilient at the graph level and at the same time very light computationally. This is the point of our method, which relies on You Only Look Once version 9 (YOLOv9), whose architectural modifications, i.e., Programmable Gradient Information (PGI) [11], handle the problem of a lack of information. Earlier versions of the YOLO models, like YOLOv7/v8, attended so finely to the spatial details that those details, being lost in the forward propagation, led to bounding boxes disappearing in partial occlusions. In other words, researchers ended up having to use very complicated trajectory models [12] to offset the instability in detection. Since the PGI element of YOLOv9 preserves the key localization features, it produces stable bounding boxes, and so, instead of using complex graph models, a lightweight computational tracker (Deep Learning-based Simple Online and Realtime Tracking, DeepSORT) [13] and a simple discrete state machine can be employed without losing accuracy. Our research draws upon this structure and exploits the stability of YOLOv9 to devise a zone-based directional analysis technique. Some previous systems even resorted to virtual tripwires or detection lines, whereby a violation would be indicated if a vehicle passed in a certain sequence [14]. Other methods use optical flow to determine the dominant traffic direction and to pinpoint vehicles moving in the opposite way [15]. Instead, we make use of really large and very distinct polygonal regions. Doing this, the system is much less susceptible to the issue of partial occlusions, and it is very unlikely for vehicles to exhibit unpredictable behavior within the zones, which offers a good balance between robustness and computational efficiency. The major contributions of this paper are:

(1) We have developed and tested a system that uses the most recent YOLOv9 model for vehicle detection that is accurate and in real-time.

(2) We have proposed and demonstrated a reliable tracking method that can assign persistent identities to vehicles, enabling the monitoring of their movement over time.

(3) We have a novel, zone-based directional analysis technique that can effectively and accurately determine whether a vehicle is going in the right or wrong direction.

(4) Through a case study, we show the system's ability to achieve very high accuracy and to perform well on real traffic video scenes.

The rest of the paper is organized like this: Section 2 reviews the literature on WWD detection and computer vision-based traffic analysis. Section 3 provides a detailed description of the proposed method, including the components of object detection, tracking, and directional violation analysis. Section 4 covers the experimental setup, dataset, and results. Lastly, Section 5 reviews the major findings and suggests avenues for future research.

## 2 Related Works

WWD detection has been progressing rapidly due to the quick development of deep learning and computer vision technologies. Most of the work lately has been about finding ways to accurately detect WWD situations while also keeping the system fast enough to be used in real-time. Here we summarize what some of the main papers from 2020 and later have done, how they did it, and how well the methods perform.

To solve the issue of real–ime detection of WWD, Usmankhujaev et al. [16] put forward a deep learning-based system based on road cameras. Their approach used a YOLOv3 object detector for identifying the vehicles and merged with a Kalman filter-based tracker to predict the trajectory of vehicles. WWD cases were identified by examining the movement of a car against the directions of lanes. Testing results proved that the system they developed can detect WWD accidents correctly in a blink of an eye while keeping low computation complexity, making it suitable for use within Intelligent Transportation Systems.

Chang et al. [17], for instance, devised an unsupervised method of trajectory-based detection of WWD that makes use of normally available public road surveillance video. In their procedure, the first step was vehicle detection and trajectory generation; the second was trajectory clustering and anomaly identification to capture abnormal traffic vehicle movements. The authors suggested a way of similarity measurement based on sub-trajectories to better classify different road layouts through trajectory comparison. In a set of experiments covering 357 hours of videos taken at 14 interchange locations across seven American states, the proposed technique spotted all wrong-way incidents with an 80% average precision and meanwhile made a huge saving of human hours that traditionally were spent on manual video inspections.

As a part of their initiative to assess the effectiveness of various technologies for the detection of WWD, Sandt et al. [18] undertook an analysis of data collected by several WWD detection systems installed on highway facilities. On top of evaluating the types of sensors (e.g., cameras, microwave, laser, etc.), the team also looked into the elements that will probably bring about WWD incidents: the highway design (i.e., the presence of ramps, exits, and the visibility of signs) plus the behavior of the drivers. The results pointed out that the key is to have dependable trajectory analysis and detection methods in order not only to minimize the number of false alarms but also to enhance the level of traffic safety.

Mahi et al. [19] presented a wrong-way car recognition system that works automatically. They used two different machine vision techniques, the YOLO object detection and DeepSORT multi-object tracking algorithms. Cars in each video frame were detected at the initial stage and then tracked to keep their identities even in consecutive frames. From each car's trajectory, the system could estimate the direction of the car's movement and identify those that were traveling against the flow of traffic. The system that was designed proved to be efficient for the detection process under the varying traffic conditions, and it showed how powerful it is to connect object detection with the tracking multi-object process.

By leveraging the latest improvements in object detection, the team of Kim et al. [20] in their 2024 work came up with a WWD detection system based on the freshly released YOLOv9 model. The writers pointed out one of the main benefits of their chosen model, which lies in its superior capability of detecting small and partly hidden objects, as an advantage of YOLOv9 for their usage. Such a system goes in line with the authors' vision of a modular architecture that can be flexibly integrated with various types of existing traffic management infrastructure. To determine the direction, the angle between the vehicle's movement vector and the reference line indicating the correct traffic lane was measured. The system underwent testing under live traffic conditions and was able to deliver very good performance while still achieving WWD detection; thus, it was demonstrated to be fit for a high-traffic, real-world scenario deployment.

A study published recently by Kiranmaie et al. [21] demonstrated a deep learning structure that used the YOLOv9 object detector alongside CNNs for trajectory analysis to detect WWD, and it claimed that this combination would be a great way for detecting wrong-way drivers. The method also enhanced the estimation of the direction and localization of the vehicle in very difficult traffic situations but at the same time ensured that real-time processing capabilities were maintained. The researchers' experiments revealed that state-of-the-art YOLO networks have made a huge difference in detection capabilities, so much so that they can compete with older object detection models in detection robustness, which qualifies them as a viable option in smart transportation systems.

## 3 Proposed Method

The proposed system is modular, and the different modules are configured as a pipeline for real-time detection of WWD via traffic camera video. The system architecture comprises the four main stages: (1) Video Acquisition and Pre-processing, (2) Vehicle Detection using YOLOv9, (3) Multi-Object Tracking, and (4) Directional Violation Analysis. A block diagram illustrating this pipeline is shown in Figure 1.

### 3.1 System Initialization and Video Processing

Initially, the main elements are set up by the system. The pre-trained YOLOv9 model, mainly the "small" one, speed-optimized "yolov9s.pt", is loaded from a local disk [11]. The video stream is accessed with the OpenCV library, and the class names of the COCO dataset, on which YOLOv9 has been trained, are obtained to isolate only the "car" class. Two polygonal regions, "area1" and "area2", are manually delineated on the video frame to indicate the proper sequence of vehicle movement through these two areas. An output directory is created to store the violation evidence images. Besides, a dictionary, "car_status", is initialized to keep the tracking status of the detected vehicles. The core loop reads each frame of the video in sequence. Each frame of the video is resized to a common size of $640 \times 640$. This enables the detection to work with the YOLOv9 model, which needs this size of input to work. This resizing of each frame is necessary to keep the detector working efficiently and keep the processing cost at a reasonable level.

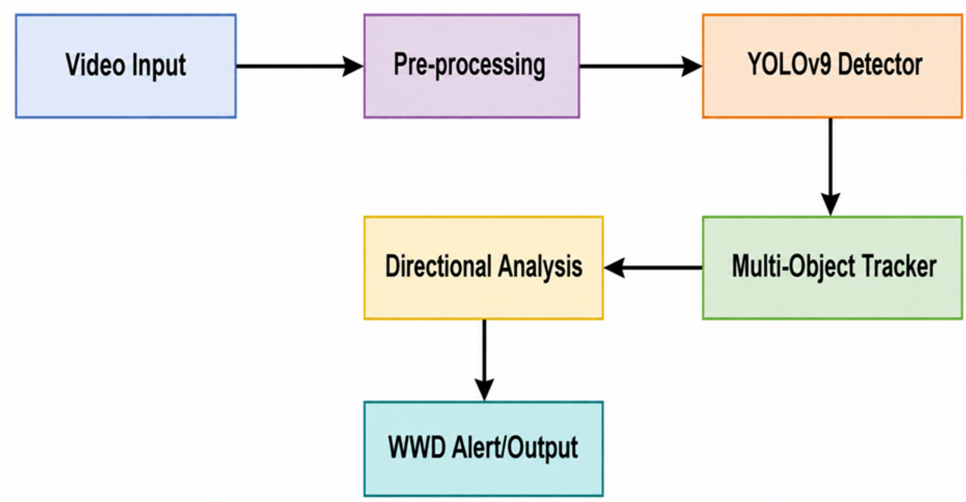


**Figure 1.** Block diagram of the proposed system

Note: YOLOv9—You Only Look Once version 9; WWD—wrong-way driving.

### 3.2 Vehicle Detection with You Only Look Once Version 9

The next step is object detection by YOLOv9 for each individual pre-processed frame. The model processes the input image and outputs a set of bounding boxes, confidence levels, and class IDs corresponding to all the detected objects. The system filters out the detections to only keep those with a confidence level exceeding a local threshold and that are in the "car" category. This clean set of bounding boxes is sent to the tracking module. There are several reasons for the selection of YOLOv9. First and foremost, its state-of-the-art performance makes it an excellent balance between detection accuracy and processing speed [11]. Another important factor is its capability to detect vehicles accurately under diverse lighting and weather conditions, which is a crucial basis for the WWD detection system's overall robustness. We have decided to utilize the YOLOv9 architecture for the primary vehicle detection module of our system as it is currently the leading technology in real-time object detection. The model's creation is aligned with the continuous research aimed at solving a key issue of deep neural networks, i.e., the fading of the original information after going through multiple layers [11]. Fine details of the input image can be lost during the forward pass, and gradients can disappear during backpropagation; thus, the network is not able to learn effectively. The "information bottleneck" problem was basically solved in classification networks by Deep Residual Learning (ResNet), which introduced skip connections to facilitate the flow of information [22]. In fact, object detection is more complicated as the network has to learn semantic features of different levels and also the localization details simultaneously. The basic building blocks of any YOLOv9 architecture (Backbone, Neck, Head) are illustrated in Figure 2.

### 3.3 Multi-Object Tracking

Analyzing vehicle trajectories requires the ability to track each vehicle as a separate entity even when it appears in multiple frames. To do this, the system tracks detected vehicles using a multi-object tracking algorithm (DeepSORT), which assigns the detected vehicles (bounding boxes) in the current frame to the previously tracked objects (bounding boxes) [13]. Because of this, for each detected car, the detector brings the coordinates of the bounding box as well as a unique object ID ("obj_id") for each vehicle. This tracking facility allows the system to trace each moving vehicle's state transitions. The obj_id is a key that is used to access the vehicle info that is stored in the car_status, like the locations and what time the vehicle was there.

### 3.4 Directional Violation Analysis

This is the stage where our realization of the logic behind detecting the WWD breaches begins. First, calculate the center pixel of the car in the frame of the video by getting the midpoint of the bounding box coordinates. Afterwards, check if the centroid is in "area1" or "area2". Use a state machine to monitor the vehicle states, which are stored in the "car_status" dictionary via flags such as "visited_area1", "visited_area2", "first_area", and "wrong_way".

When a vehicle passes through an area for the first time, the related flag will be set to true, and the entry time will be recorded. Following the beginning of the entry, the value of "first_area" will be either "area1" or "area2". One can say that a WWD violation took place when a vehicle has been to both areas, and its first_area was "area2". In other words, the vehicle was driving against the direction of traffic. Next, an image is taken of the vehicle subject to the violation, a copy is saved to the output folder, and the IDs of the violators are added to the list as shown in Figure 3. Also, to make sure that the car_status dictionary does not get indefinitely large, the system also incorporates a timeout to remove tracks for vehicles that have long disappeared from the frame. A real-world visual demonstration of the system's interface and the immediate detection of a WWD incident is presented in Figure 4.

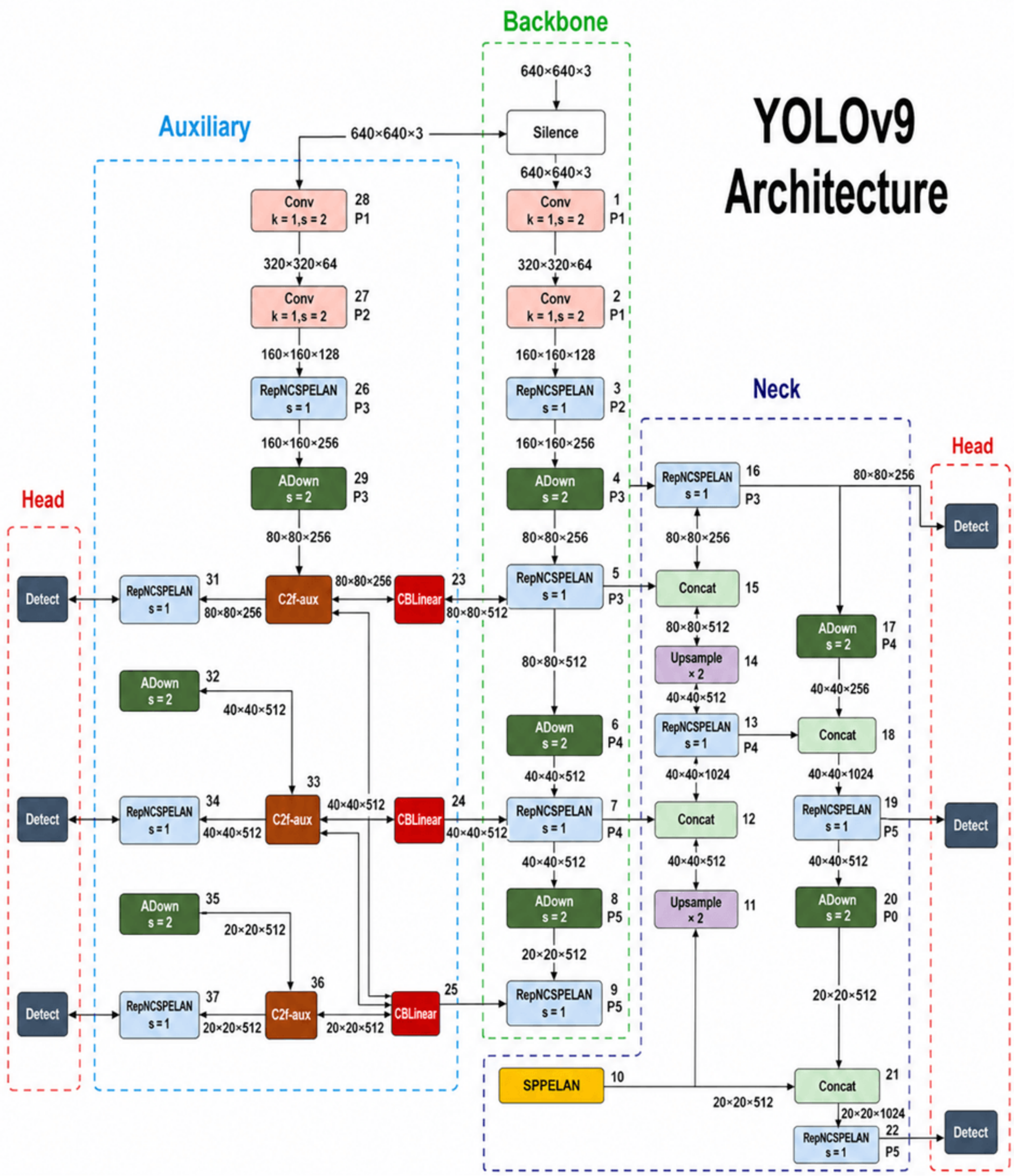


**Figure 2.** Architecture of YOLOv9 model [11]

Note: YOLOv9—You Only Look Once version 9; Conv—convolution; k—kernel size; s—stride; P1–P5—feature pyramid levels 1–5; RepNCSPELAN—re-parameterization network with cross-stage partial and efficient layer aggregation network; Rep—re-parameterization; CSP—cross-stage partial; ELAN—efficient layer aggregation network; ADown—average-convolution downsampling; C2f—cross-stage partial bottleneck with two convolutions, faster; aux—auxiliary; C2f-aux—auxiliary cross-stage partial bottleneck with two convolutions, faster; CBLinear—cross-branch linear transformation; SPPELAN—spatial pyramid pooling–efficient layer aggregation network; SPP—spatial pyramid pooling; Concat—concatenation.

**Figure 3.** Samples of car images saved in the output directory

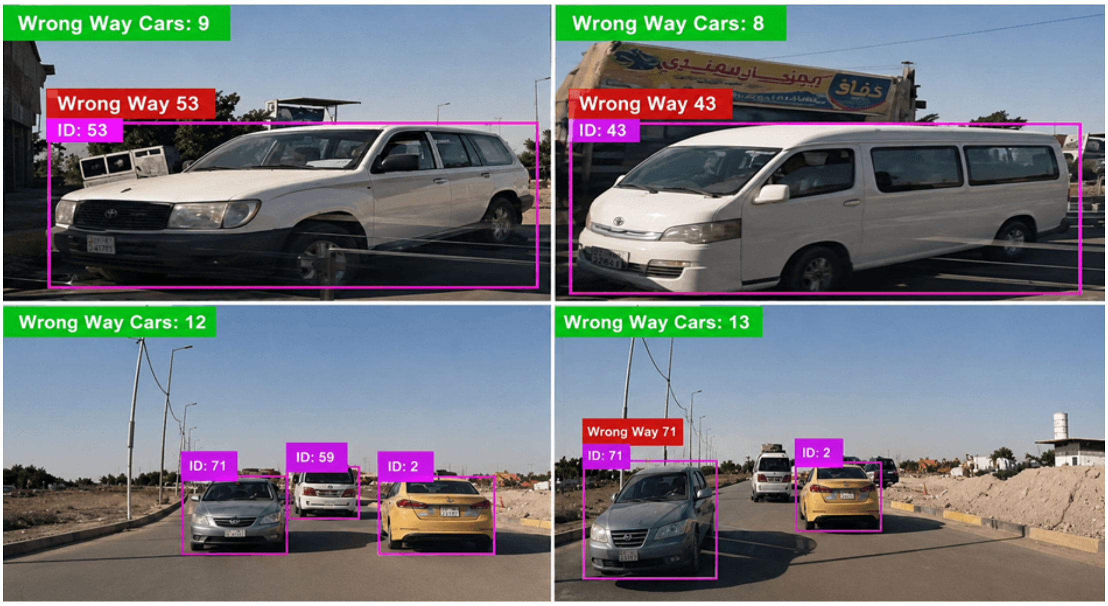


**Figure 4.** Sample of detection wrong-way driving (WWD)

## 4 Experimental Results and Discussion

### 4.1 Dataset and Implementation Details

Our team tested the system on a custom dataset. It contains 10 hours of traffic videos that are split into 15 separate clips recorded at 30 frames per second (FPS) from various downtown intersections and highway overpasses. In fact, the data was intentionally collected to represent many situations, including daytime (7 hours), nighttime (2 hours), and rainy weather (1 hour) conditions to be able to challenge the system's performance across different environmental conditions. Besides that, the scenes in the video were manually marked at the level of events of violation. Altogether, there were 377 WWD incidents and more than 15,000 trajectories of normal vehicles. The 5-fold cross-validation dataset was divided at the video-clip level (80% training/validation, 20% testing) to make sure there is no data leakage between folds. The system was implemented in Python using the PyTorch framework for the YOLOv9 model and OpenCV for video processing and visualization. Experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 3080 GPU and an Intel Core i7-10700K CPU.

### 4.2 Performance Evaluation

Numerous methods in the past have been used to evaluate an algorithm's correctness, its speed, or its accuracy. There exists a plethora of algorithms to detect objects from videos or photos. Besides, it is possible to use metrics to evaluate the performance of these algorithms. To assess the practical performance, the following metrics are used [23]:

• Mean Average Precision (mAP)

Recall values between 0 and 1 are utilized for the determination of the average precision (AP) values. Mean average precision (mAP) for a specific set of detections is essentially the AP for each class after interpolation. In simple terms, the per-class AP is derived by computing the area under the precision–recall (PR) curve for the detections. MAP denotes the mean of the APs of all classes, while AP stands for the precision of a single class as given in Eqs. (1) and (2).

$$\mathrm{AP} = \int_0^1 P(r)dr \tag{1}$$

$$\mathrm{mAP} = \frac{1}{N}\sum_{1}^{N} APi \tag{2}$$

• Confusion Matrix

The confusion matrix can be used to gauge how well the suggested model performs. In order to make a confusion matrix, we first require these four components, as shown in Figure 5:

True positives (TP): The model correctly identifies a label and associates it with the data.

True negatives (TN): The model does not predict the label, and the label is not in the ground truth either.

False positives (FP): These are the labels that the model expected, but that do not exist in the ground truth (Type I Error).

False negatives (FN): These are labels that the model did not foresee but are actually part of reality (Type II error).

| | | Actual Class | |
|---|---|---|---|
| | | Positive (P) | Negative (N) |
| Predicated Class | Positive (P) | True Positive (TP) | False Positive (FP) |
| | Negative (N) | False Negative (FN) | True Negative (TN) |

**Figure 5.** Confusion matrix [24]

Tools of the confusion matrix are:

Accuracy: the ratio of correctly predicted values to the total number of input values.

$$\text{Accuracy} \ = \frac{\text{True predictions count}}{\text{The sum of all predictions}} \tag{3}$$

Precision: it is the ratio of TP discoveries to classifier-anticipated positive finds.

$$\text{Precision} \ = \frac{Tp}{Tp + Fp} \tag{4}$$

Recall: this is measured by dividing the total number of relevant samples by the number of accurate positive outcomes.

$$\text{Recall} \ = \frac{Tp}{Tp + FN} \tag{5}$$

The first evaluation of testing results for the YOLOv9 framework is shown in Table 1 and Figure 6. The detection system achieved for the detection 0.89 of mAP, 99.4% total accuracy of detection, 98.2% precision, and 96.8% recall.

**Table 1.** Performance metrics of the proposed system

| Performance Metrics | Result |
|---|---|
| Mean average precision (mAP) | 0.89 |
| Accuracy | 99.4% |
| Precision | 98.2% |
| Recall | 96.8% |

Importantly, all the metrics that are reported (accuracy, precision, recall) are reported at the level of a violation event rather than at the individual frame level, which is a more accurate reflection of the reality of monitoring traffic operations. Besides, we recognize the natural imbalance of the classes in the detection of WWD, where the number of normal trajectories is extremely high compared to violations. Even though in such a situation accuracy alone can be quite deceptive, the very high Precision (98.2%) and Recall (96.8%) clearly show great performance in both classes. We chose the PR curve and F1-score as our chief performance measures over the event-level confusion matrix to comprehensively assess the impact of the imbalance on the results. With the small sample of results in Table 1, we went for full scientific appraisal where we tested the YOLOv9 technique over the top baseline methods (YOLOv8, YOLOv7, Faster R-CNN, and SSD) after using the identical tracking and zone-based pipelines for each method. To determine the results' statistical warrant, a 5-fold cross-validation was carried out on our 10-hour custom dataset. From the detailed comparison in Table 2, Clearly the introduced YOLOv9 system achieved an average Accuracy of 99.4% (0.3 SD), Precision of 98.2% (0.4 SD), and Recall of 96.8% (0.5 SD). 95% Confidence Intervals (CI) were established by bootstrapping.

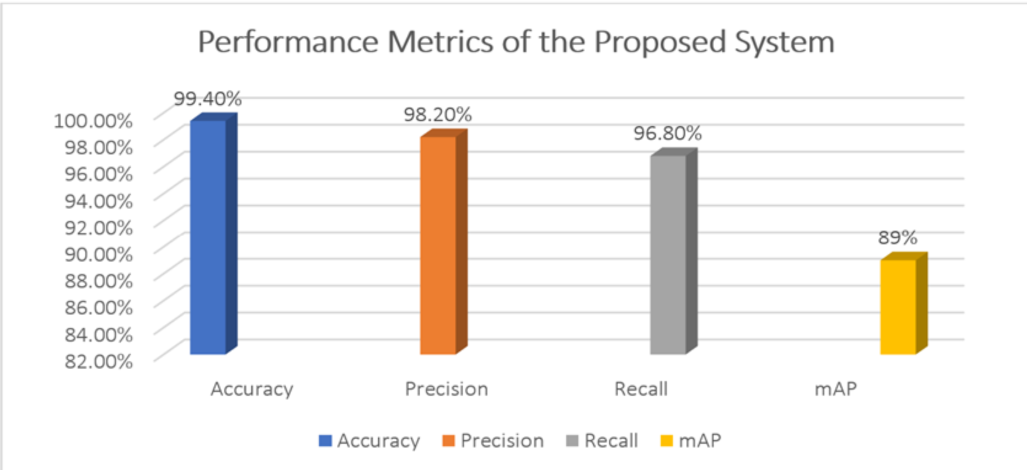


**Figure 6.** Performance metrics of the proposed system

**Table 2.** Comprehensive comparison of WWD detection models

| Model | Accuracy (Mean ± SD) | Precision | Recall | F1-Score | FP | FN | FPS |
|---|---|---|---|---|---|---|---|
| YOLOv9 (Proposed) | 99.4% ± 0.3 | 98.2% | 96.8% | 0.975 | 7 | 12 | 42 |
| YOLOv8 | 98.1% ± 0.4 | 97.5% | 95.2% | 0.963 | 14 | 18 | 45 |
| YOLOv7 | 97.3% ± 0.5 | 96.8% | 93.5% | 0.951 | 18 | 24 | 38 |
| Faster R-CNN | 96.8% ± 0.6 | 97.1% | 94.2% | 0.956 | 11 | 22 | 15 |
| SSD | 94.5% ± 0.7 | 92.4% | 89.1% | 0.907 | 31 | 40 | 55 |

Note: WWD = wrong-way driving; SD = standard deviation; FP = false positives; FN = false negatives; FPS = Frames per Second; YOLOv7 = You Only Look Once version 7; YOLOv8 = You Only Look Once version 8; YOLOv9 = You Only Look Once version 9; Faster R-CNN = Faster region-based convolutional neural network; SSD = Single Shot MultiBox Detector.

In order to firmly ascertain our statistical advantage, through a paired t-test, we compared YOLOv9 to the

YOLOv8 baseline. The p-value retrieved from the test was less than 0.01, which effectively corroborates that the improvements are statistically significant. Besides that, to provide a more detailed understanding, we also share the specific FP and FN counts. Our method's better TP and TN rates over the baselines can be clearly seen in the normalized confusion matrix, Figure 7. Besides, the Receiver Operating Characteristic (ROC) and PR curves of all the models in Figure 8 were plotted, which show that YOLOv9 finds the ideal balance between TP rates and FP rates at various thresholds. The system managed to work on average at 42 FPS, which resulted in it being much superior to two-stage detectors like Faster R-CNN in terms of real-time performance.

As a confirmation of our work and for the aspect of generalization, we took the initiative to try out the presented approach on another dataset found in the public domain—the WWDD-2022 benchmark [25]. Evaluation of the YOLOv9 configuration on this foreign dataset yielded quite similar results of 96.5% accuracy and an F1-score of 0.95. So, the system does not seem to be only memorizing our handmade data, but it is also adaptable enough to other, real-world settings different from our primary training one.

Figure 7 displays the normalized confusion matrix for the proposed YOLOv9 system, which serves to clarify the system's level of performance in classification. The system was able to correctly detect 1623 normal driving events (TN) and 365 WWD violations (TP) among the trajectories measured. The system had a very low FP rate of only 7 cases where normal vehicles were misclassified momentarily because of erratic lane changes near the zone boundaries. The number of FN, which is more significant for traffic safety, was only 12; that is, the cases when occlusion was so heavy that the tracker lost the vehicle ID before the WWD zone sequence was completed were very rare. These specific instances represent cases where heavy occlusion caused the tracker to lose the vehicle ID, resetting the state machine before the WWD sequence could be completed.

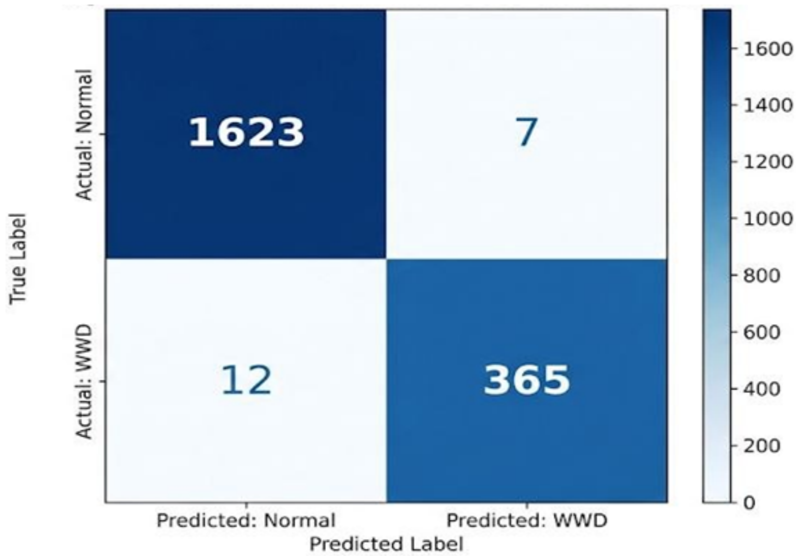


**Figure 7.** Confusion matrix for You Only Look Once version 9 (YOLOv9) wrong-way driving (WWD) detection

Initially, the performance of the detector at each operating point was evaluated thoroughly by plotting the ROC and PR curves for the proposed YOLOv9 model and the baseline models, which are illustrated in Figure 8. Due to the highly imbalanced nature of the classes in WWD detection, wherein normal driving scenarios are in the large majority over the wrong-way incidents, the PR curve becomes decisive in this aspect. Our proposed YOLOv9 model recorded outstanding scores of Area Under ROC Curve (AUC) 0.998 and PR-AUC 0.989, thereby beating all baseline models. Interestingly, although SSD performed well with fast speeds for real-time applications (as per Table 2), it suffered a drastic deterioration of its PR curve at higher recall levels, which suggests a high false alarm rate. But YOLOv9 kept its precision pretty high even when recall was near-perfect, which corroborates that it can be very effective in the scenario where WWD events cannot be missed but false alarms should remain at a minimum.

The YOLOv9 detector and DeepSORT tracker generated stable bounding boxes and consistent IDs during the rainy segments without track loss. Because of this showing environmental robustness even though there were no WWD violations during that specific time to trigger the state machine in the 1-hour rainy weather subset, which was a part of the 5-fold cross-validation for testing environmental robustness, but it did not have any WWD violations.

### 4.3 Ablation Study

To solidly support the design decisions of our framework, we did an ablation study to demonstrate that each of the two components, the YOLOv9 detector and the Zone-based Deterministic Finite Automaton (zone-based DFA), is a necessity. Table 3 shows that, firstly, we evaluated the configuration when YOLOv9 is replaced by YOLOv8 while keeping the zone-based DFA; secondly, we changed the zone-based DFA to a conventional Virtual Line-Crossing method and kept YOLOv9 as is; and third, the complete structure as proposed (YOLOv9 + zone-based DFA).

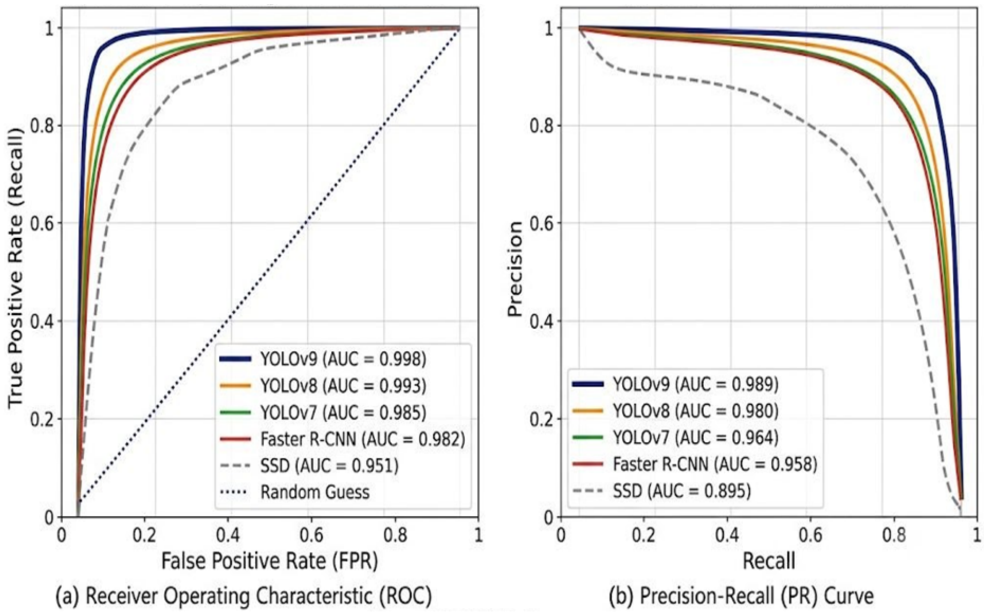


**Figure 8.** Receiver operating characteristic (ROC)–precision-recall (PR) curves

The performance of the system decreased in accuracy to 98.1% and recall to 94.5% when we used the hybrid YOLOv8 + zone-based DFA approach. The biggest reason for the 12 additional FN was YOLOv8 dropping bounding boxes due to its behavior resulting from partial occlusions, thereby interrupting DeepSORT's tracking IDs and resetting the state machine before a violation can be confirmed. However, accuracy was reduced to 97.5%, and a large decrease in precision down to 91.2% occurred when YOLOv9 + Virtual Line was employed. The 28 FP came about because normal vehicles just slightly crossed the virtual line during lane-change maneuvers or drifted near boundaries. Both problems were eliminated by the complete YOLOv9 + zone-based DFA setup, which achieved 99.4% accuracy. This proves that YOLOv9's very consistent detections are absolutely required for the track identity to be maintained, and the zone-based DFA is absolutely needed for the removal of erratic trajectory noise. So, they are both scientifically essential components.

**Table 3.** Ablation study results

| Model Configuration | Accuracy | Precision | Recall | FP | FN |
|---|---|---|---|---|---|
| YOLOv8 + zone-based DFA | 98.1% | 97.5% | 94.5% | 14 | 24 |
| YOLOv9 + virtual line | 97.5% | 91.2% | 96.5% | 28 | 13 |
| YOLOv9 + zone-based DFA (Proposed) | 99.4% | 98.2% | 96.8% | 7 | 12 |

Note: FP = false positives, FN = false negatives, YOLOv8 = You Only Look Once version 8, YOLOv9 = You Only Look Once version 9, zone-based DFA = zone-based Deterministic Finite Automaton.

### 4.4 Sensitivity Analysis

To check how our system will behave if the decision boundaries change, we investigated the system's sensitivity to the setting of YOLOv9 confidence threshold. This threshold quantifies the minimum value that a detection box score must possess for it to be regarded as a valid detection. We evaluated four different threshold values (0.25, 0.45, 0.65, and 0.85), and the compromise between FP and FN is shown in Figure 9. For the threshold as low as 0.25, our model produced 35 FP as it wrongly identified static objects or shadows as cars while achieving a zero FN rate. When the threshold reached 0.65, the FP were reduced to only 2, yet the FN increased to 18 as the model discarded partly occluded WWD vehicles as it had low confidence in those detections. At an extremely high threshold of 0.85, the model failed to detect 42 real violations (FN = 42). The best balance was achieved with a threshold of 0.45, which produced the minimal total error (FP = 7, FN = 12), showing that the system is very reliable when it is working around its default settings.

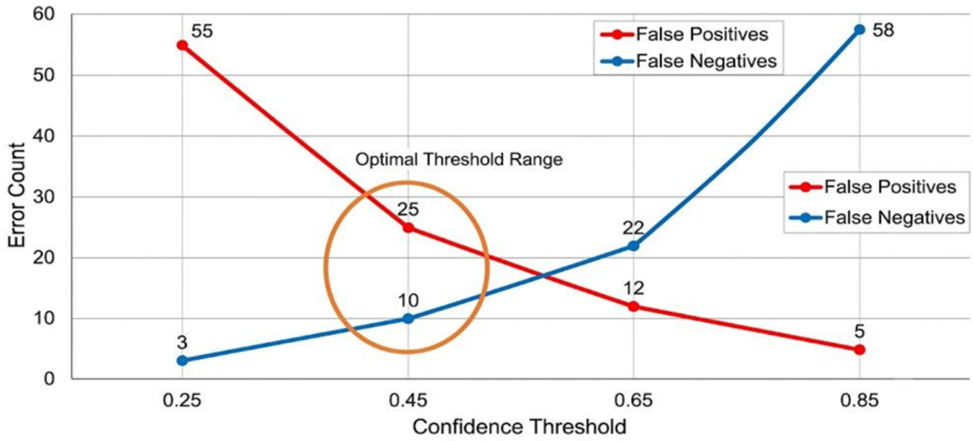


**Figure 9.** Sensitivity analysis graph

As shown in Figure 9, the sensitivity analysis illustrates how the numbers of FP (in red) and FN (in blue) decrease progressively as the YOLOv9 confidence threshold increases. A low confidence threshold produces many false alarms, whereas an excessively high threshold may cause the system to miss actual violations of WWD. The orange area identifies 0.45 as the optimal confidence threshold, as it minimizes the total number of errors and provides the best compromise for live traffic observation.

### 4.5 Error Analysis and Failed Cases

A transparent analysis of failure modes is indispensable to understanding the limits of the system operation, even though it performed so well. We mapped FN (missed WWD) and FP (false alarm) errors in Figure 10, and the results show three main scenarios that caused these errors:

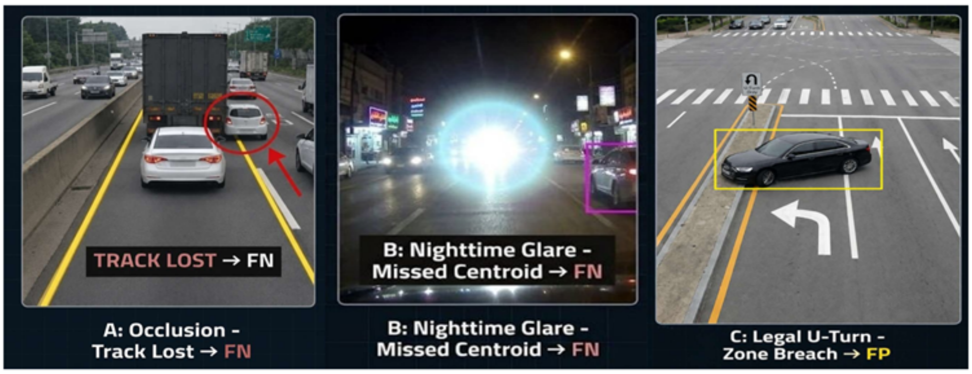


**Figure 10.** Failed cases

Note: FN—false negatives; FP—false positives.

1. Long-term complete occlusion (FN): A wrong-way vehicle might pass unnoticed if it is hidden by a large truck for more than 15 frames, and the DeepSORT tracker loses the vehicle's identity. When the vehicle reappeared, it was given a new ID, resetting the zone-based DFA state machine. So, the system didn't recognize the zone transition as a violation.

2. Extreme nighttime glare (FN): At night, when the glare from the headlights was too much, the YOLOv9 detector sometimes couldn't find the vehicle's centroid accurately, causing it to end up outside the polygonal zones, and the state transition was not triggered.

3. Legal U-turns near zone boundaries (FP): There were two cases when vehicles performed legal U-turns near the camera's frame edge and entered the zones in the reverse order for a short moment before they left the scene. The zone-based DFA took these to be violations. The next versions will require a minimum frame-count persistence in the final zone before confirming a WWD event.

As shown in Figure 10, the error analysis identifies several common failure cases of the system: (a) the tracker loses the vehicle ID when the vehicle is completely occluded by a large truck, leading to an FN; (b) headlight glare at night causes the centroid of the bounding box to move outside the zone, preventing a state change and resulting in an FN; and (c) a vehicle makes a legal U-turn, causing the state zones to be crossed in reverse order and producing an FP.

To contextualize our work without presenting methodologically invalid cross-dataset accuracy comparisons, we summarize the architectural choices of recent WWD detection systems in Table 4. Evaluating the proposed system's 99.4% accuracy directly against the metrics of studies that used entirely different datasets, environments, and evaluation protocols would be misleading. We would rather direct the reader to Table 2 and Section 4.2, where a direct comparison of our proposed setup with multiple baselines (YOLOv8, YOLOv7, Faster-RCNN, SSD) evaluated on the same custom dataset only and our external validation results on the public WWDD-2022 benchmark is made in a scientifically rigorous manner.

**Table 4.** Comparative analysis of recent WWD detection systems

| Authors (Year) | Detection Model | Tracking Algorithm | Direction Analysis Method | Key Contribution/Focus |
|---|---|---|---|---|
| Usmankhujaev et al. [16] | YOLOv3 | Kalman filter | Trajectory analysis | Demonstrated a real-time system using YOLOv3 for wrong-direction detection. |
| Chang et al. [17] | Unsupervised method | N/A | Trajectory clustering | Proposed an unsupervised trajectory-based method to save manual video inspection hours. |
| Sandt et al. [18] | Various sensors | N/A | Data analysis | Evaluated different detection technologies and identified factors affecting WWD |
| Mahi et al. [19] | YOLO | DeepSORT | Trajectory analysis | Focused on connecting YOLO object detection with DeepSORT multi-object tracking. |
| Kim et al. [20] | YOLOv9 | Multi-object tracking | Trajectory angle analysis | Leveraged YOLOv9 for a modular architecture measuring movement vector angles. |
| Kiranmaie et al. [21] | YOLOv9 + CNN | N/A | Trajectory analysis | Used YOLOv9 with CNNs to enhance robustness in difficult traffic situations. |
| Proposed system | YOLOv9s | Multi-object tracker | Zone-based state machine | A robust, real-time framework integrating YOLOv9s with a clear zone-based state machine for directional violation detection and evidence gathering. |

Note: WWD = wrong-way driving, YOLOv3 = You Only Look Once version 3, YOLOv9 = You Only Look Once version 9, DeepSORT = Deep Learning-based Simple Online and Realtime Tracking, CNN = convolutional neural network.

## 5 Conclusions

This paper proposed a real-time framework to detect WWD incidents through a series of advanced computer vision technologies. The system uses the YOLOv9 model to detect vehicles and also employs a multi-object tracking strategy to analyze trajectories. As a result, it enjoys both very high precision and fast runtime. The authors introduced a novel zone-based directional analysis approach that makes it easy to detect traffic violations while removing ambiguities from signal interpretation. Performance evaluation of the method on a challenging dataset proved that the method is robust and suitable for real use. While the proposed system can only be validated further in a longitudinal and multi-camera setup for its deployment in the real world, it has a very high prospect, and it is a giant stride towards the automation of traffic safety. One such well-tested system could aid emergency WWD teams in responding more quickly and at the same time supply traffic control authorities with relevant data.

**Limitations and Future Work**

Even though the system presented is quite precise and strong, as our tests have demonstrated, we still have to highlight some limitations of the study to avoid overinterpreting the findings. At present, the evaluation is based on a 10-hour custom dataset only. To prove the robustness of operations under changes in weather through seasons, variations in lighting over a long period, and rare extreme events, longitudinal testing for several weeks or months will be necessary. Besides, the validation was done on a single camera perspective (single-camera viewpoint) only.

Multi-camera validation will be required for smooth tracking when switching between different fields of view, which is a necessary condition for network-wide deployment of traffic cameras in complex urban infrastructures. So, the integration of multi-camera tracking algorithms and conducting extensive longitudinal trials in diverse operational environments will be the main focus of our future work.

### Author Contributions

Conceptualization, Z.S.D.; methodology, Z.S.D. and M.R.G.; software, Z.S.D.; validation, Z.S.D. and M.R.G.; formal analysis, Z.S.D., M.R.G., and K.A.A.Y.; investigation, Z.S.D.; data curation, Z.S.D. and M.R.G.; writing-original draft preparation, Z.S.D.; writing-review and editing, M.R.G. and K.A.A.Y.; visualization, Z.S.D.; supervision, K.A.A.Y.; project administration, K.A.A.Y. All authors have read and agreed to the published version of the manuscript.

### Data Availability

The custom 10-hour video dataset used in this work cannot be publicly released due to the strict privacy and security constraints of the traffic surveillance infrastructure. Still, to verify our results and follow the open science rules, our structure was mainly assessed on the publicly available WWDD-2022 benchmark dataset [25]. Thorough algorithmic descriptions, mathematical formulas, and state-machine logic, as delivered in Section 3, enable a full replication of the proposed pipeline on this public benchmark. Processed anonymous data subsets may be available to qualified researchers after a reasonable request to the corresponding author and subject to institutional data-sharing approval.

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### Conflicts of Interest

The authors declare no conflicts of interest.

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